

Comprehensive Project Blueprint: Academic Publishing, Education & Services Platform

Architecture: Headless (Laravel 13 API + Next.js SSR)

Infrastructure: Hostinger KVM-2 VPS + AWS S3 Storage + MySQL 8

1. Introduction

This document outlines the complete technical and developmental roadmap for a full-scale, bespoke Academic Publishing Platform. This system is designed as a unified, single-domain ecosystem that handles rigorous academic peer review, public knowledge dissemination, institutional collaborations, and premium commercial services.

By leveraging a decoupled headless architecture, the platform guarantees lightning-fast page loads for maximum Search Engine Optimization (SEO), while simultaneously providing secure, highly interactive backend dashboards for authors, reviewers, and platform administrators. The platform includes a complete Manuscript Tracking System (MTS), an Institutional Hub for universities, an educational Course Hub, an Events Portal, and a Professional Services portal.

2. System Architecture & Database Strategy

To ensure enterprise-grade scalability and cost-efficiency without relying on complex multi-tenant environments, the system utilizes a centralized architecture:

- **The Backend Engine (Laravel 13):** Acts purely as an API. It handles the complex relational database logic, state machines for manuscript routing, user authentication, and automated email queues.
- **The Frontend Interface (Next.js):** Consumes the Laravel API. It uses Server-Side Rendering (SSR) for all public-facing pages (Blogs, Journals, Archives) to ensure perfect indexing by Google Scholar.
- **Storage Optimization (AWS S3):** Because the system runs on a Hostinger KVM-2 VPS, all heavy media—manuscripts, course videos, podcasts, and payment receipts—are piped directly to AWS S3. This keeps the primary server lightweight and prevents disk overflow.
- **MySQL Database Architecture:** The platform uses a highly normalized MySQL 8 database. Because we are not using isolated multi-tenancy, the database utilizes relational linking (e.g., mapping an `institution_id` to users). This allows the platform to centrally host all journals, while allowing registered Universities to track the published papers of their associated students and faculty.

3. Comprehensive Feature Matrix

3.1 Institutional Hub & Memberships (B2B Features)

- **University Registration & Portals:** Universities can register an organizational account. Admins can generate specific invite links or domain-whitelists (e.g., @harvard.edu) to onboard their students.
- **Student Publishing Access:** Students associated with a registered University gain fast-tracked Author access, and their published papers are aggregated on a dedicated University Profile Page.
- **Article Processing Charge (APC) Waivers:** Logic allowing institutions to pre-pay publishing quotas so their students bypass manual APC fees at checkout.

3.2 Public Portals, CMS & Media

- **Headless Blogging & Podcasts:** Full SEO capabilities, rich text editing, category/tag routing, and multimedia embeds for podcasts. Integrated Google AdSense slots for revenue generation.
- **Events Portal:** A directory for upcoming academic conferences, webinars, and summits, complete with registration dates and external links.
- **Public Journal Archives:** Hierarchical browsing system navigating by Journal -> Publication Year -> Volume -> Issue -> Individual Article.
- **Abstracting & Indexing Ready:** Automated metadata generation (XML feeds) to easily push published articles to PubMed, Scopus, CABI, and Google Scholar.
- **Advanced Global Search:** Server-side search engine spanning all articles, blogs, events, and courses.

3.3 Manuscript Tracking System (MTS) Core

- **Strict State Machine Workflow:** Enforces the academic lifecycle: *Draft -> Submitted -> Technical Triage -> Under Review -> Revisions Required -> Accepted -> Published*.
- **Author Dashboard:** A secure portal for researchers to submit new papers via a multi-step wizard, manage co-authors, and track manuscript status in real-time.
- **Editorial Board Workflows:** Dedicated interfaces for Editors-in-Chief to view submissions, run plagiarism checks, and assign Associate Editors.
- **Reviewer Dashboard & Routing:** A secure portal for invited academics to accept/decline review requests, securely download manuscripts, and submit standardized evaluation scorecards.

3.4 E-Commerce & Verification Engine

- **Course Management System (LMS):** An educational hub featuring video lessons, structured modules, and progress tracking.
- **Professional Services Portal:** A system for users to request premium editorial and

translation services.

- **Manual Payment Verification System:** A highly secure checkout flow where users (or institutions) view platform bank details, upload a PDF/Image of their wire transfer receipt, and await Admin approval to unlock their product or publish their manuscript.

3.5 Security & Integrity

- **Automated Plagiarism Detection:** Direct API integration with Copyleaks to automatically scan submitted manuscripts for duplicated content and AI-generated text.
- **Granular Role-Based Access Control (RBAC):** Strict security policies ensuring users can hold multiple concurrent roles (e.g., Author, Reviewer, and Student) without accessing unauthorized data.

4. Detailed Implementation Phases & Task Breakdown

Phase 1: Environment, Database Architecture & Security

Establishment of the server infrastructure, MySQL relations, and API security protocols.

Task	Detailed Description	Est. Hours
Server Provisioning	Configure the Hostinger KVM-2 VPS. Install Ubuntu, Nginx web server, PHP 8.3-FPM, MySQL 8, and Node.js. Setup PM2 for Next.js.	6h
Core API & S3 Integration	Initialize Laravel 13 framework. Configure Laravel Flysystem to connect with AWS S3 for direct frontend uploads.	8h
MySQL Relational Schema	Architect and migrate the MySQL database tables for Users, Institutions, Roles, Journals, Volumes, Issues, Articles, Submissions,	14h

	Reviews, and Courses. Include strict Foreign Key constraints.	
RBAC Implementation	Build Laravel Middleware to restrict API endpoint access based on roles (Super Admin, Editor, Associate Editor, Reviewer, Author, Student/Institution).	10h
Authentication System	Implement Laravel Sanctum for secure token generation. Build Next.js authentication contexts for login, registration, and secure route protection.	12h

Phase 2: CMS, Blogging, Events & SEO Foundation

Building the public-facing content engine to drive organic traffic.

Task	Detailed Description	Est. Hours
CMS API Development	Build Laravel CRUD endpoints for dynamic Pages (About, Guidelines), Blog Posts, and Podcast links.	12h
Events Portal API	Develop the backend structure for academic Events (Title, Dates, Location, Call for Papers deadline,	8h

	Registration links).	
Admin WYSIWYG Editor	Integrate a React-based rich text editor into the Admin dashboard. Configure it to upload embedded images directly to AWS S3.	10h
Next.js Public Routing	Develop the Next.js frontend pages for Blogs, Events, and static pages using Server-Side Rendering (SSR).	16h
Technical SEO Implementation	Code dynamic XML sitemaps, JSON-LD structured data for Google Rich Snippets, and inject Google AdSense script components into layouts.	10h

Phase 3: E-Commerce, Courses & Manual Verification Engine

Developing the commercial portals and the offline receipt-approval workflow.

Task	Detailed Description	Est. Hours
Course Engine API	Build backend logic for Course creation, Module/Lesson structuring, and User Enrollment status tracking.	12h

Course Player UI	Develop the Next.js student dashboard. Includes a video player interface, lesson navigation sidebar, and progress tracking.	16h
Service Booking Workflow	Create a multi-step frontend form for users to request "Editorial Services," upload their raw documents to S3, and generate invoices.	12h
Manual Checkout System	Build the checkout UI displaying platform bank details. Create a secure upload component for users to submit their transaction receipts.	10h
Admin Approval Ledger	Develop a backend dashboard for Administrators to view all pending orders, view the uploaded receipt, and trigger an "Unlock" action.	14h

Phase 4: Institutional Hubs & Academic Archives

Engineering the B2B university portals and public research display.

Task	Detailed Description	Est. Hours
Institutional Registration	Backend and frontend logic	14h

Flow	for Universities to apply for a profile. Includes generating unique registration codes for their student bodies.	
University Profiles UI	Public Next.js pages displaying a University's logo, bio, and an aggregated feed of all articles published by their students/faculty.	12h
Academic Hierarchy APIs	Create highly optimized, read-only Laravel endpoints that deliver nested data: Journals -> Volumes -> Issues -> Articles.	12h
Journal Archives UI	Build the Next.js pages for browsing Journal history and Editorial Boards.	16h
Article Reading View	Develop the single-article UI displaying abstracts, authors, affiliations, citations, and secure PDF downloads.	16h

Phase 5: Manuscript Tracking System (MTS) Core

The core engineering phase: building the private portals for academic peer review.

Task	Detailed Description	Est. Hours

MTS State Machine Backend	Code the rigid logic governing manuscript transitions. Log all state changes in a MySQL history table.	16h
Author Dashboard (Next.js)	Develop a complex, multi-step React form. Step 1: Metadata. Step 2: Co-author invites. Step 3: Direct-to-S3 file uploads. Step 4: Live tracking.	24h
Editor-in-Chief Dashboard	Build the command center. View incoming submissions, verify Institutional affiliations, and assign Associate Editors.	20h
Reviewer Assignment System	Create the logic for Editors to search the user database for qualified reviewers and trigger secure email invitations.	18h
Reviewer Dashboard	Develop the secure portal for active reviewers. Include access to the specific manuscript PDF and a dynamic React Scorecard form.	18h

Phase 6: Integrity Tools, Automated Communications & Deployment

Finalizing platform integrity, system notifications, and production launch.

Task	Detailed Description	Est. Hours
Copyleaks API Integration	Write a background Laravel Job that automatically sends newly submitted manuscripts to the Copyleaks API and attaches the report to the Editor's dashboard.	16h
Asynchronous Email Queues	Configure Laravel Queues to handle heavy email tasks. Integrate the Resend API to fire automated alerts (e.g., "Receipt Approved," "Review Deadline").	14h
End-to-End System QA	Comprehensive testing simulating a full journey: University registration, student signup, manuscript submission, peer review, and publication.	24h
Server & DB Optimization	Configure Redis for caching heavy API database queries. Set up Nginx FastCGI caching. Configure server SSL certificates via Cloudflare.	10h

5. Technology Stack & Third-Party Operating Costs

The platform utilizes the following technologies to maintain a lean operating budget while

delivering premium performance.

Tool / Provider	Platform Function	Estimated Monthly Cost
Hostinger KVM-2	Primary VPS hosting the MySQL Database, Backend API, and Frontend App.	~\$8.00 - \$10.00 / month
AWS S3	Cloud storage bucket for all heavy files, preventing VPS disk limits.	~\$1.00 - \$3.00 (Pay-as-you-go)
Resend API	Transactional email provider for all automated system notifications.	Free (up to 3,000 emails/mo)
Copyleaks API	Artificial Intelligence and Plagiarism detection scanner.	~\$10.99 per 25k words
Zoho Mail	Professional human inboxes (e.g., admin@yourdomain.com).	Free (up to 5 users)
Cloudflare	DNS Management, Global CDN routing, and SSL Security.	Free Tier

6. Project Summary & Timeline

- **Total Estimated Development Effort:** ~446 Hours.
- **Estimated Timeline:** 11 to 14 weeks of dedicated, full-time development.
- **Conclusion:** By utilizing a centralized MySQL database architecture combined with an

Institutional tracking system, this platform achieves the scale and collaborative power of an enterprise system without the developmental overhead of multi-tenancy. The integration of manual payment verification, educational courses, and a robust Manuscript Tracking System ensures the platform is fully equipped to generate immediate revenue while establishing long-term academic authority.